

connected between Vcc and pin 3 through current-limiting resistor R5. When the circuit is powered, LED3 turns on and glows continuously. Here only the output stage of the IC is used.

Experiment 2

Circuit diagram for the second experiment is shown in Fig. 2. Here the IC is wired as a monostable multivibrator.

A blue-coloured LED4 is connected between resistor R7 and capacitor C1. When the circuit is powered, LED4 glows as it gets

grounded through the internal discharge transistor at pin 7. However, capacitor C1 cannot charge. When S1 is pushed momentarily, the capacitor starts charging through R7 and LED4.

As the capacitor starts charging, it pulls the charge through LED4 and the LED starts fading as the voltage across it drops. When the capacitor voltage exceeds two-third of Vcc, it discharges through pin 7 because of internal comparators as per normal monostable operation and LED4 glows again.

Though output pin 3 is not used, the monostable operation of the circuit is visible through the fading and glowing of LED4.

Experiment 3

The circuit diagram for experiment 3 is shown in Fig. 3. Here the IC is wired as an astable multivibrator so that the two LEDs, LED5 and LED6, flash and fade alternately.

Capacitor C2 charges through resistor R8 and LED6 and discharges through LED5 and resistor R9, causing the LEDs to flash and fade alternately. The flashing and fading of the LEDs show the charging and discharging of capacitor C2, which occurs because of the IC's internal comparators as per normal astable operation. LED5 and LED6 are 5mm, transparent bright-green LEDs.

Thus, without connecting any load to output pin 3 of the IC, its astable operation can be observed.

Experiment 4

Circuit diagram for the fourth experiment is shown in Fig. 4. Here the IC is wired as a bistable multivibrator.

In this circuit, the LED is connected between Vcc and pin 7 of the IC through current-limiting resistor R12. In conventional bistable operation, the discharge pin 7 of IC 555 is not used as the circuit has no capacitor.

Initially, when the circuit is powered, LED7 turns on. When switch S2 is pushed, LED7 turns off, and when switch S3 is pushed, it turns on. This is just the reverse of the conventional bistable operation as instead of output pin 3, discharge pin 7 is used.

In this circuit the function of the internal discharge transistor (pin 7) can be visibly seen through LED7. Normally, pin 7 of the IC is used to discharge the capacitor when the IC is operating as an astable or a monostable multivibrator.

Experiment 5

Circuit diagram for experiment 5 is shown in Fig. 5. Here two ICs, IC7 and IC8, are used. IC7 is wired as a monostable multivibrator and IC8 is wired as a bistable multivibrator.

Discharge pin 7 of both the ICs are connected together. Initially, when the circuit is powered, the output at pin 3 of IC7 and IC8 is low and both LED8 and LED9 are off.

When switch S4 is pushed, LED8 turns on and stays on continuously, though the circuit is wired as a monostable multivibrator. This is because capacitor C3 is shorted to ground by the internal discharge transistor pin 7 of IC8. Due to this capacitor C3 cannot charge and the monostable operation fails to occur.

In the same situation, when

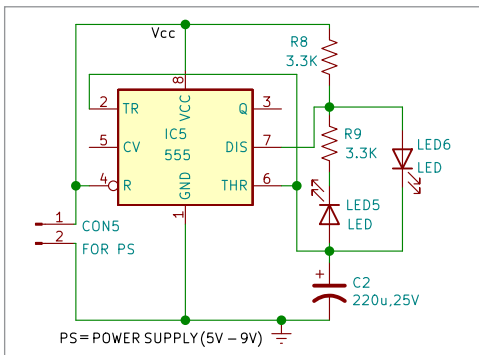


Fig. 3: Circuit diagram for experiment 3

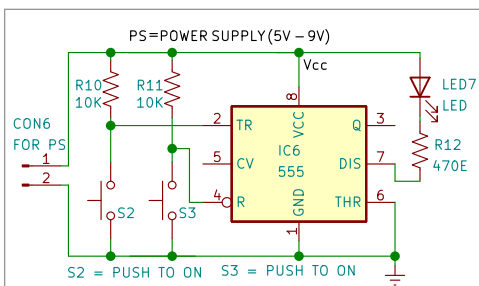


Fig. 4: Circuit diagram for experiment 4

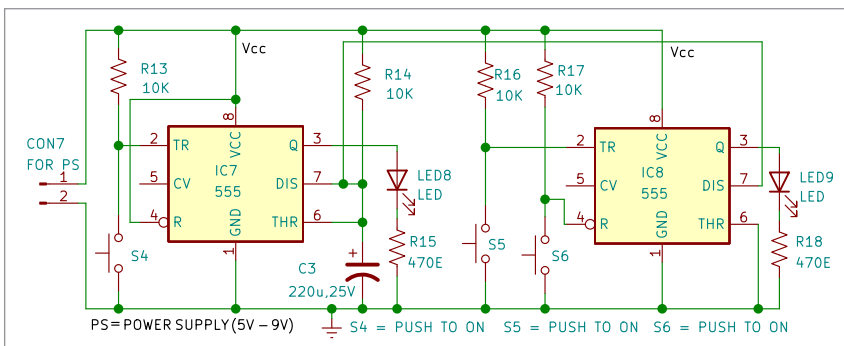


Fig. 5: Circuit diagram for experiment 5